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Effects of Karting Competition on Heart Rate Variability, Fatigue, Neuromuscular Function, and Cognitive-Motor Performance in Racing Drivers: An Exploratory Study

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Abstract

Background/Objectives: Competitive karting places high physiological and cognitive demands on drivers. This field study investigated the acute effects of racing on heart rate variability (HRV), perceived fatigue and neuromuscular function, and cognitive-motor performance during an official event held under persistent rain. Methods: Longitudinal, repeated-measures design across two conditions: control (race video viewing) and competition (qualifying and race). Four drivers (Junior, X30 Senior, X30 Super Shifter) were assessed pre/post-control and post-qualifying/race on Day 1 (Saturday) and pre/post-race only on Day 2 (Sunday). Measures included continuous heart rate, pre/post HRV (HRV4Training; rMSSD, SDNN), perceived fatigue (ROF), bilateral handgrip strength, and visuomotor performance (reaction times and accuracy). Results: On Day 1, SDNN and rMSSD decreased significantly after qualifying versus pre- and post-control ($p < 0.05$), remaining globally lower post-race; no changes emerged in frequency-domain indexes. Perceived fatigue, handgrip strength, and mean/max reaction times did not change significantly; an improvement in minimum reaction time was observed post-race versus post-control ($p = 0.033$). rMSSD consistently decreased after racing on both days ($p < 0.05$) with no day × time interaction observed, and accuracy improved on Sunday, reflected by more correct attempts (hits) and fewer failed attempts (errors) ($p < 0.05$). Conclusions: Racing was associated with lower time-domain HRV indices (rMSSD/SDNN), consistent with heightened autonomic activation without measurable decrements in handgrip-based neuromuscular function or cognitive-motor performance. The improved accuracy on Day 2 may be related to an increased level of physiological activation and readiness associated with race day. Routine HRV monitoring across race weekends is recommended to guide recovery decisions when subjective scales show limited immediate sensitivity. These findings are preliminary due to the small and heterogeneous sample and should be interpreted cautiously.

Keywords: karting; motor racing; motorsport; heart rate variability; fatigue; visuomotor performance; BlazePod; handgrip strength

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1. Introduction

The performance of motorsport drivers is influenced by a complex combination of physical, psychological, and environmental factors [1]. During karting competitions, drivers are exposed to extreme temperature conditions and high physical stress [2], resulting in imbalanced variations in physiological markers. These variations may result in increased

fatigue, negatively affecting performance and compromising essential cognitive functions. Central fatigue models, which propose fatigue originates from central nervous system mechanisms rather than peripheral muscle fatigue alone [3], may be particularly relevant in karting. The sustained cognitive demand, combined with physical stress and environmental factors, could lead to neurotransmitter imbalances (e.g., serotonin/dopamine) or altered cortical arousal—mechanisms central to these models—ultimately impairing reaction time, decision-making, and consistent performance throughout the race [4].

Although some studies have explored physiological responses in motorsport, there is still limited research on how heart rate variability (HRV), fatigue, and cognitive-motor performance interact in real competitive settings [4,5]. Most of the literature focuses on higher categories, simulators, or controlled laboratory environments, which do not fully reproduce the demands of competitive karting. This gap is particularly important since karting often involves young athletes who are in the process of physiological and psychological development, which may distinctly influence their response to race-induced stress [4].

Karting represents a distinct area of motorsports that needs further evidence. It is widely recognized as the entry point to competitive motorsport and the environment in which most elite drivers acquire their foundational skills, often under conditions of considerable physical and cognitive demand. Unlike higher racing categories, karting vehicles lack effective suspension, power steering, and thermal protection. As a result, drivers are directly exposed to substantial physical stressors, including high-frequency vibration and sustained lateral G-forces that are otherwise partially mitigated at higher levels [6]. In addition, karting competitions typically consist of multiple short, high-intensity heats conducted within a single event. This structure imposes repeated bouts of effort separated by limited recovery, creating conditions in which fatigue accumulates across heats and recovery remains incomplete [7]. Beyond these physical factors, karting takes place in a highly immersive competitive environment characterized by emotional pressure, radiant heat from the engine, high noise levels, and continuous tactical interaction with opponents. Such conditions cannot be fully reproduced using simulators or laboratory-based protocols. Taken together, these characteristics position karting as a valuable real-world model for examining the interplay between fatigue, cognitive-motor performance, and physiological responses in a genuine high-performance and high-risk sporting context.

From a physiological perspective, heart rate (HR) is one of the most widely used parameters to monitor exercise intensity. Studies in motorsport have shown that drivers' HR during competition can reach high values, often close to 80–90% of maximal heart rate, revealing an internal load comparable to that of high-intensity aerobic disciplines [8]. This response results from the combination of physical effort required to control the vehicle—particularly under lateral G-forces and intense vibration—and sympathetic autonomic nervous system activation triggered by competitive stress. While HRV monitoring has proven valuable for assessing training load and recovery in youth athletes from various sports [9], its application to understanding the acute psychophysiological stress of motorsport competition, particularly in its foundational discipline, remains scarce.

HRV is another relevant indicator, allowing for the analysis of autonomic nervous system activity. Marked reductions in indices such as SDNN and rMSSD during competition reflect sympathetic predominance and reduced parasympathetic influence, suggesting a high psychophysiological load [10]. Monitoring HRV in young drivers therefore becomes a valuable tool for understanding how they respond to competitive effort and for assessing their capacity to recover between different stages of the race [5].

In parallel, motorsport requires high cognitive performance. Drivers must process information within fractions of a second, anticipate opponents' behavior, manage race

strategies, and react quickly to visual stimuli [4]. The literature shows that experienced drivers demonstrate advantages in sustained attention and reaction time compared to untrained individuals [11]. However, both physical and mental fatigue can impair these functions, leading to slower responses and a greater likelihood of attentional lapses [12].

Reaction time, in this context, is a key variable. In karting races, minimal differences can determine the effectiveness of an overtaking maneuver or the ability to avoid an accident. However, reaction time alone does not fully capture the complexity of cognitive demands in racing [13]. Inhibitory control—the ability to suppress prepotent but contextually inappropriate responses—is equally critical, as drivers must continuously override automatic impulses to adapt to dynamically changing track conditions and opponent behaviors [14]. Technological tools such as BlazePod™ have been increasingly used to practically and reliably assess reaction speed and cognitive-motor performance in sports contexts and may offer a practical means of monitoring potential performance changes associated with competitive fatigue [15].

Understanding how HRV, perceived fatigue, neuromuscular function, and cognitive-motor performance fluctuate over the course of a race [5], and how these variables relate to competitive performance, may support the development of strategies aimed at enhancing performance and driver safety [16,17]. This knowledge provides a foundation for designing targeted physical and mental preparation frameworks for karting, as well as implementing adaptive monitoring and recovery strategies aligned with drivers' developmental and experiential profiles.

In this context, the present exploratory study focuses on karting, a sport of high physical and cognitive demand from an early age, aiming to help fill the existing knowledge gap regarding physiological and neuromuscular responses in real competitive environments. The study aims not only to analyze the acute effects of a karting race but also to compare how these responses change across the different evaluation moments and between the race and control (film) conditions. The main objective is to examine the changes in heart rate variability (HRV), neuromuscular function, perceived fatigue, and cognitive-motor performance, and to understand how these variables vary over the assessment points and across both days.

2. Results

For the present study, seven drivers were invited to participate. Of these, five expressed interest in taking part. Two drivers did not agree, citing race preparation as a priority and expressing concern that participation in the study could compromise their competitive focus. One driver withdrew after the initial assessment for the same reason. The final sample consisted of four karting drivers (Table 1). The participants were male, with a mean age of 21.8 years and an average competitive experience of 7.5 years. Regarding anthropometric characteristics, the drivers presented a mean height of 170 cm and an average body weight of 60 kg.

Table 1. Characterization of the Participant Sample.

Participant	Age (Years)	Height (cm)	Weight (kg)	Karting Experience (Years)
1	12	160	45	3
2	14	168	50	7
3	16	172	55	10
4	45	180	90	10

Table 2 presents the results of the measurements taken throughout the first day of competition, which included the viewing of the film, participation in the qualifying heat,

and the final race. No significant differences were observed over time in the variables of perceived fatigue, neuromuscular function assessed through maximal voluntary contraction, or cognitive-motor performance expressed by maximum, minimum, and mean reaction time, correct attempts, and failed attempts. On the other hand, significant differences over time were detected in the time-domain variables of HRV, namely rMSSD ($p = 0.017$) and SDNN ($p = 0.022$). Specifically, rMSSD values were significantly lower after qualifying compared with those recorded after the control condition ($p = 0.037$). Although SDNN values decreases after both qualifying and race relative to the post-control condition, and rMSSD values were also lower after the race compared with the post-control condition, differences were not significant after Bonferroni correction.

Table 2. Physiological Response Throughout the First Day of Competition.

Variable	Pre-Control (M1)	Post-Control (M2)	Post-Qualifying (M3)	Post-Race (M4)	*p*-Value
Fatigue	3.5 (1.5–4.8)	5.0 (2.0–5.8)	3.0 (2.3–5.3)	4.5 (4.0–6.5)	0.227
Right Upper Limb MVC	32.8 (21.8–51.7)	31.1 (22.3–42.7)	32.4 (23.1–48.6)	32.0 (21.9–44.5)	0.347
Left Upper Limb MVC	31.5 (21.9–44.1)	30.3 (22.7–42.3)	31.7 (23.2–42.4)	30.1 (21.6–40.9)	0.132
SDNN	48.8 (44.5–55.6)	54.1 (39.2–65.1)	28.0 (21.6–34.2)	32.5 (19.1–40.6)	**0.022**
rMSSD	46.2 (35.2–54.5)	50.1 (36.2–65.5)	22.7 (15.4–27.1) [a]	26.2 (18.0–29.8)	**0.017**
Reaction Time (mean)	529.0 (461.3–611.8)	547.5 (477.3–629.0)	515.0 (443.3–612.3)	521.5 (470.5–611.5)	0.098
Reaction Time (max)	971.0 (835.8–1293.0)	1072.0 (850.3–1293.5)	921.5 (741.8–1438.0)	918.5 (831.0–1273.8)	0.943
Reaction Time (min)	380.0 (351.5–462.5)	380.0 (350.5–405.0)	371.0 (323.3–406.8)	345.6 (330.5–371.0)	0.086
Correct Attempts	49.0 (37.8–53.5)	42.0 (36.3–48.5)	48.0 (38.5–52.3)	44.0 (35.5–50.3)	0.335
Failed Attempts	7.5 (3.8–10.5)	9.5 (8.3–12.3)	9.0 (7.3–10.8)	9.0 (5.5–14.0)	0.417

Legend: MVC, Maximum Voluntary Contraction; rMSSD, root mean square of successive differences between normal RR intervals; SDNN, standard deviation of all normal RR intervals. Values are presented as median (percentile 25–75). Significance level: $p < 0.05$. [a] significantly different from post-control condition, $p = 0.037$.

Table 3 presents the comparative results during Saturday (Day 1) and Sunday (Day 2) of competition. No significant differences were observed between days in the variables of perceived fatigue, handgrip strength (right and left), and HRV (SDNN and rMSSD). In the HRV domain, SDNN and rMSSD values decreased after race in both race days, but differences were not significantly different. Regarding cognitive-motor performance, no changes were observed in mean and maximal reaction times, but there was a borderline difference in the fastest reaction times ($p = 0.059$), where the quickest reaction times appear to have improved after the race on Saturday ($p = 0.056$). A significant difference was also detected in cognitive-motor performance accuracy: drivers achieved more correct attempts before and after race on Sunday compared with Saturday; however, these differences were not significant after Bonferroni adjustment.

Table 3. Physiological Response Between Two Consecutive Days of Competition.

Variable	Pre-Race Saturday (M1)	Post-Race Saturday (M2)	Pre-Race Sunday (M3)	Post-Race Sunday (M4)	*p*-Value
Fatigue	5.0 (2.0–5.8)	4.5 (4.0–6.5)	3.0 (2.3–3.8)	4.0 (3.0–6.5)	0.290
Right Upper Limb MVC	31.1 (22.3–42.7)	32.0 (21.9–44.5)	32.8 (21.2–46.0)	31.9 (22.1–46.4)	0.960
Left Upper Limb MVC	30.3 (22.7–42.3)	30.1 (21.6–40.9)	34.0 (23.4–44.9)	31.7 (20.5–43.5)	0.094
SDNN	54.1 (39.2–65.1)	32.5 (19.1–40.6)	54.1 (29.6–72.2)	32.1 (24.2–34.4)	0.144
rMSSD	50.1 (36.2–65.5)	26.2 (18.0–29.8)	50.4 (26.9–78.7)	26.0 (18.3–34.0)	0.273
Reaction Time (mean)	547.5 (477.3–629.0)	521.5 (470.5–611.5)	496.0 (446.8–591.8)	469.5 (460.3–659.5)	0.348
Reaction Time (max)	1072.0 (850.3–1293.5)	918.5 (831.0–1273.8)	1056.0 (773.5–1220.5)	928.0 (687.8–2917.3)	0.825
Reaction Time (min)	380.0 (350.5–405.0)	345.6 (330.5–371.0)	360.0 (334.5–377.3)	350.5 (337.5–433.3)	0.059
Correct Attempts	42.0 (36.3–48.5)	44.0 (35.5–50.3)	49.5 (44.3–58.5)	52.5 (40.8–56.8)	**0.034**
Failed Attempts	9.5 (8.3–12.3)	9.0 (5.5–14.0)	5.5 (4.0–7.8)	5.5 (4.0–7.8)	0.070

Legend: MVC, Maximum Voluntary Contraction; rMSSD, root mean square of successive differences between normal RR intervals; SDNN, standard deviation of all normal RR intervals. Values are presented as median (percentile 25–75). Significance level: $p < 0.05$.

3. Discussion

The present exploratory study analyzed, in a real competitive context, the effects of a karting race weekend on cardiovascular variables (i.e., HRV), neuromuscular function (through handgrip strength), perceived fatigue, and cognitive-motor performance. On the first day of competition, significant changes were observed in the time-domain HRV variables, namely SDNN and rMSSD. Specifically, rMSSD values were lower after qualifying.

In addition, there were differences in cognitive-motor performance accuracy across race days, where accuracy seemed to improve on the second day of racing (more correct attempts).

Heart Rate Variability:

The observed reduction in rMSSD following competitive sessions, particularly after qualifying, is indicative of a shift toward sympathetic predominance and reduced self-regulation. In motorsport contexts, such reductions in time-domain HRV indices are commonly reported during competition and are understood to arise from the combined influence of sustained cognitive demand, emotional stress, and physical loading, including vibration exposure and prolonged isometric muscle activation [5,8,18–20].

In the present study, this autonomic response was evident on both competition days, suggesting a consistent effect of the competitive context itself rather than day-specific fluctuations. Importantly, the wet-track conditions may have further amplified psychophysiological stress through increased task uncertainty, heightened attentional demands, and the need for continuous ongoing task adaptation, even in the presence of reduced absolute lateral forces.

The persistence of reduced rMSSD values post-race, without parallel increases in perceived fatigue or decrements in neuromuscular performance, supports the interpretation that the observed HRV changes primarily reflect acute competitive arousal and stress-related autonomic activation rather than cumulative physiological fatigue. Nevertheless, given the short PPG-based recording windows, official competition constraints, and the small and heterogeneous sample, these findings should be interpreted as preliminary and hypothesis-generating.

Perceived fatigue and neuromuscular function: The absence of significant changes in perceived fatigue and handgrip strength suggests that, within the immediate post-race assessment window, the specific demands of this event did not elicit detectable neuromuscular impairment. This finding should be interpreted in light of the officially declared wet-race conditions, which are typically associated with reduced tire–asphalt friction, lower cornering speeds, and consequently diminished lateral G-forces. From a biomechanical perspective, such conditions may reduce sustained isometric loading on the forearm musculature and steering grip, potentially attenuating peripheral fatigue development.

In parallel, competitive arousal may have contributed to a transient attenuation of fatigue perception. Acute sympathetic activation and motivational drive are known to modulate central fatigue mechanisms, allowing individuals to maintain subjective alertness and functional readiness despite underlying physiological strain. This phenomenon aligns with central fatigue models emphasizing supraspinal regulation of perceived effort during high-arousal tasks [21,22].

Importantly, the use of validated instruments (ROF scale and digital handgrip dynamometry) and standardized testing procedures support the reliability of the measurements. However, the lack of observable neuromuscular decline should not be interpreted as evidence of the absence of fatigue but rather as an indication that any fatigue-related effects may have been masked by competitive arousal, mitigated by race conditions, or occurred outside the temporal resolution of the assessment protocol.

Cognitive-motor performance: On the first day of competition, the reduction in minimum reaction time observed after the race, relative to the control condition, may reflect a transient facilitation of response speed associated with increased physiological and cognitive activation. This pattern is consistent with arousal-based models of performance, in which moderate elevations in central activation can acutely enhance reaction speed without necessarily impairing response accuracy [23,24]. Importantly, this effect was confined to the fastest responses, suggesting an improvement in peak responsiveness rather than a global shift in the overall reaction-time distribution [25,26].

Across competition days, accuracy was higher on Sunday, with a greater number of correct responses and fewer errors. This improvement may reflect increased task familiarity, stabilization of attentional control, or short-term adaptation to repeated exposure to both the competitive environment and the visuomotor task [27]. Such adaptations may contribute to a more efficient balance between arousal and executive control during the second day of competition.

However, these findings must be interpreted with caution. The observed effects did not remain statistically significant after correction for multiple comparisons, and the very small and heterogeneous sample substantially limits statistical power, increasing susceptibility to both Type I and Type II errors. Also, environmental stressors such as rain, temperature fluctuations, and noise can modulate cognitive performance in athletes. However, evidence linking these environmental factors to cognitive decline in motorsport, including karting, remains limited and inconclusive [4]. Consequently, the present results should be regarded as exploratory and hypothesis-generating rather than confirmatory. Nonetheless, the consistent direction of the observed changes, namely faster minimal reaction times without concomitant accuracy decrements and improved accuracy on the second day—suggests that competitive karting may acutely modulate cognitive-motor performance through mechanisms related to arousal, readiness, and task engagement rather than fatigue-induced impairment.

Practical implications: The results from the present study support HRV monitoring (rMSSD, SDNN) throughout the race weekend to track activation levels and guide recovery between sessions (e.g., slow breathing, sleep hygiene, individualized periodization), particularly when subjective effort scales show limited changes. The acute improvement in reaction speed and higher accuracy on the second day suggest that warm-ups that moderately elevate arousal before decisive sessions may be beneficial. Specific cognitive-motor training (focused on inhibitory control and selective attention), when performed through tasks closely resembling competitive conditions (reactive stimuli, target selection with distractors, time pressure), can enhance transfer to on-track performance, contributing to faster reaction times, lower response variability, and fewer errors without compromising decision stability [13,28].

Limitations and future directions: This study presents several limitations that should be considered when interpreting the findings. The small sample size (n = 4) substantially reduces statistical power and limits generalizability; therefore, the present results should be regarded as preliminary evidence. In addition, the sample displayed marked heterogeneity in age and anthropometric characteristics. Although age-related autonomic differences may influence absolute HRV values, this does not compromise the main objective of the study, which focuses on within-driver physiological changes across the competitive timeline rather than direct comparisons between individuals.

HRV was also assessed using one-minute recordings obtained via smartphone photoplethysmography (PPG). This method was selected because it is non-invasive, time-efficient, and well-suited to the operational demands of a real competitive setting. While PPG offers lower temporal resolution than multi-lead ECG, independent validation has shown very

high agreement between HRV4Training and multichannel ECG for rMSSD, positioning it among the most accurate smartphone-based HRV tools [29]. Thus, under the logistical constraints of race conditions, this approach provided a robust and practical solution for short-term HRV monitoring.

Additionally, operational demands during the immediate post-race period limited both the duration and breadth of physiological monitoring. Future studies should explore longer recording windows and incorporate multi-parameter monitoring—such as internal and external load, sleep, and perceived stress—and consider age-homogeneous samples or stratified analyses to further refine the characterization of drivers' physiological responses and recovery profiles.

Specific control of fatigue-modifying substances such as caffeine was not feasible due to the constraints of the official competition setting. As caffeine intake may influence autonomic and perceptual responses, this represents an additional methodological limitation that should be addressed in future studies.

The competitive setting inherently introduces variability (race dynamics, pressure, environmental conditions), which is difficult to control yet represents the ecological reality of motorsport performance. Future designs may benefit from integrating additional controlled sessions—such as a non-competitive practice day—to isolate learning effects and reduce contextual variability. Similarly, incorporating repeated familiarization trials for cognitive-motor tasks would help minimize practice-related improvements, which could not be implemented here due to logistical constraints.

Future studies could integrate physiological monitoring with onboard telemetry data (e.g., speed, steering input, braking forces, or lateral acceleration) to better link autonomic and cognitive responses with real-time driving demands, thereby enhancing ecological validity. Future research should also explore the potential associations between physiological, cognitive, and neuromuscular markers and performance outcomes in more controlled environments, where key performance-determining variables can be manipulated or standardized. This would allow a clearer understanding of how fatigue, arousal, and cognitive processing contribute to driving performance and safety.

4. Materials and Methods

Study Design: The present exploratory study is an experimental trial with a longitudinal design, consisting of repeated measures under control, qualifying, and race conditions. The longitudinal follow-up covered two consecutive competition days (Saturday and Sunday) within the same race weekend. Each participant was assessed at different time points during the National Karting Championship, held at the Viana do Castelo circuit. Accordingly, the primary analyses focused on within-subject pre/post comparisons across assessment moments, which reduce—although do not eliminate—the potential confounding influence of age on HRV outcomes.

On the first day of the race weekend (Saturday), all participants (or their legal guardians, in the case of minors) were informed about the objectives and procedures of the study and signed informed consent prior to participation, in accordance with the principles of the Declaration of Helsinki. The study was approved by the Ethics Committee of the University of Maia (277/2025). A clinical anamnesis was conducted to collect health history information, followed by an anthropometric assessment. Participants then underwent a familiarization period with the procedures and assessment instruments to ensure adaptation to the study conditions.

After completing the familiarization phase, participants were evaluated before and after the control condition, during which they watched a video of a karting race in which they did not participate as drivers. The video was shown on the same day and at the

competition venue, in a quiet and controlled room adjacent to the paddock tent and had a duration and characteristics similar to those of official races, ensuring greater comparability between conditions. This control condition was selected to provide attentional engagement and time-matched exposure without the physical load of driving, thereby enabling within-subject comparisons of pre- and post-condition autonomic and performance measures. Following the control condition, participants were evaluated before and after two official karting events: a qualifying session and a race. On the second day of the race weekend (Sunday), participants were again assessed before and after an official race with characteristics identical to those of the previous day's event.

This study adopted an exploratory and pioneering nature within a real competitive context. The sample size was limited by operational constraints of the event (very short time windows between sessions, multiple categories, and regulatory priorities), which prevented the inclusion of a larger number of participants. In contrast, a design including a control condition was adopted to increase analytical sensitivity given the available sample, without disrupting the competitive flow.

No direct performance outcomes were analyzed, as race results in karting are highly context-dependent and influenced by multiple external factors, which could confound the interpretation of physiological and cognitive responses.

Participants: The sample consisted of karting drivers who competed in Viana do Castelo in the first round of the 2025 National Karting Championship, comprising the X30 Junior, X30 Senior, and X30 Super Shifter categories. Participants were considered eligible if they were registered in the championship within the target categories. Drivers who reported the use of ergogenic or pharmacological agents that could interfere with fatigue values, heart rate (HR) behavior, or cognitive function were excluded.

Assessments: Before and after the control condition, as well as after the qualifying session and the race on the first day of competition (Saturday), and before and after the race on the second day of competition (Sunday), data were collected regarding HR variability (HRV), perceived fatigue, neuromuscular function, and cognitive-motor performance. The assessment timeline is illustrated in Figure 1. Due to the official schedule, the assessment performed immediately after the control video coincided with the pre-qualifying window (Post-Control = Pre-Qualifying). Likewise, the assessment immediately after qualifying coincided with the pre-race window (Post-Qualifying = Pre-Race). For clarity, time points are described by both their label and contextual meaning throughout the manuscript.

Additionally, general information from the 24 h prior to the race (e.g., food intake, engagement in vigorous physical activity, presence of symptoms, consumption of caffeinated or alcoholic beverages, pharmacological agents, and tobacco use) was recorded to control for potential moderating factors, as specified in the data collection form.

Heart Rate Variability

HRV was assessed before and after each condition using the HRV4Training mobile application (Android version, A.S.M.A. B.V., Amsterdam, The Netherlands), validated for ultra-short-term HRV measurements via PPG through the smartphone's optical sensor (camera and flash) [30,31]. Participants remained seated in silence for 2 min, followed by a validated 1 min measurement using the app, from which the following indicators were obtained: mean HR, SDNN, rMSSD, and pNN50. This methodology has demonstrated acceptable agreement with ECG-derived HRV indices and heart rate monitor–based systems in athletic and applied sport settings [30,32–34].

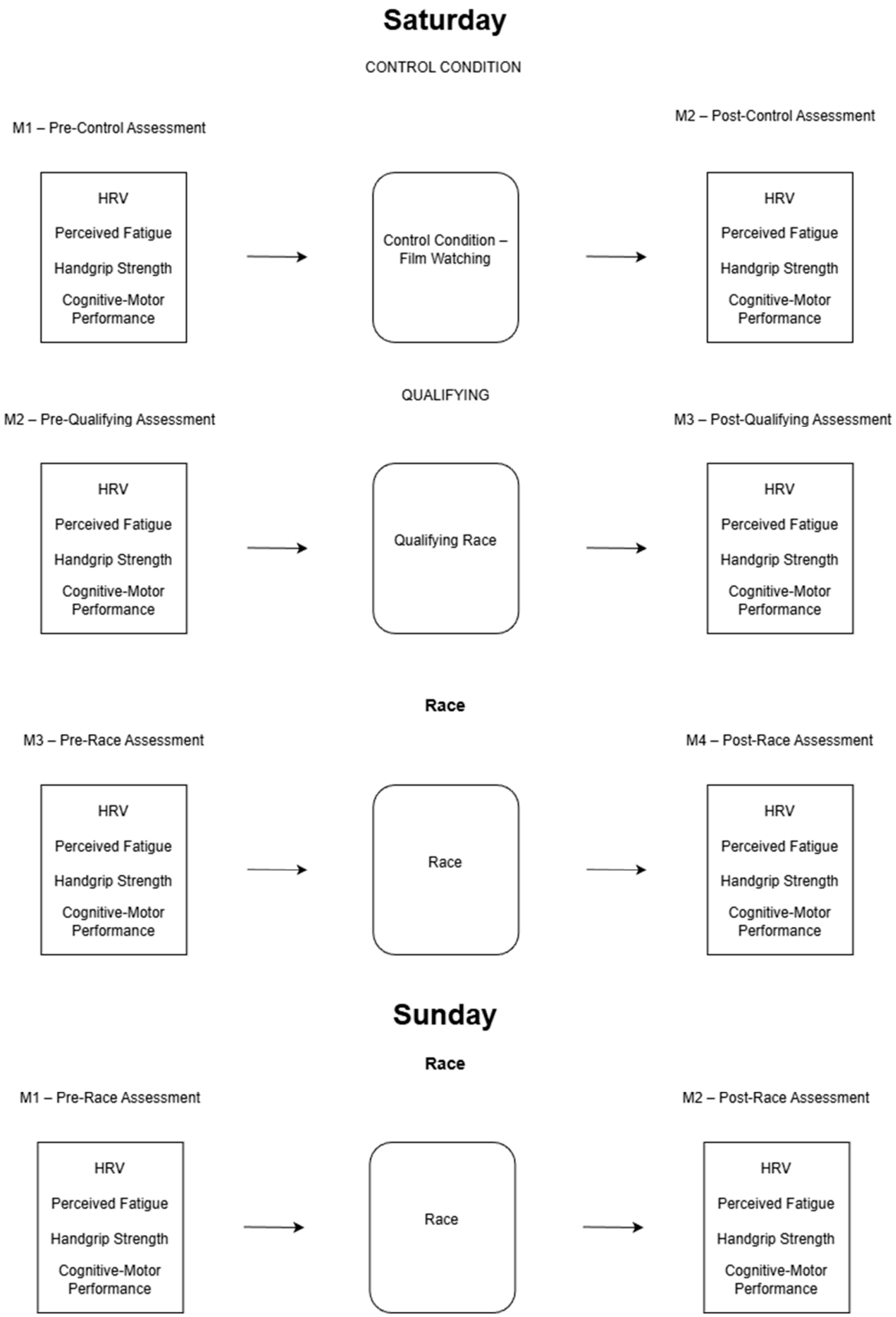


Figure 1. Design of the study. M2 corresponds to both Post-Control and Pre-Qualifying assessments. M3 corresponds to both Post-Qualifying and Pre-Race assessments.

Although standard short-term HRV assessment guidelines typically recommend 5 min ECG recordings under controlled laboratory conditions, the implementation of such procedures was not compatible with the practical demands of the competitive environment examined in this study. Between race sessions, participants were required to engage in immediate post-race performance-related activities, including technical feedback, telemetry analysis, and discussions with team staff aimed at optimizing subsequent on-track performance. The time-sensitive and cognitively demanding nature of this post-race phase limited the feasibility of prolonged physiological assessments requiring extensive preparation, immobility, or repeated sensor placement. In this context, a rapid, non-invasive HRV assessment approach was adopted to allow standardized data collection while minimizing disruption to essential performance-analysis processes intrinsic to competitive karting.

Perceived Fatigue: Subjective fatigue was assessed using the Rating of Fatigue Scale (ROF) [16], validated for the Portuguese population [35]. The scale ranges from 0 (absence of fatigue) to 10 (extreme exhaustion), with numerical anchors reflecting increasing levels of perceived fatigue.

Participants were instructed to respond promptly with a single number that best reflected their current level of fatigue, avoiding double responses or hesitation. The scale also includes qualitative descriptors and small diagrams to facilitate comprehension of each fatigue level.

The administration of this scale followed the original guidelines of Micklewright and co-authors [16] regarding its validity and consistency. At each assessment point (before and after each condition), the investigator presented the scale to the participant and requested, "Please indicate your current level of fatigue," after which the value reported by the driver was recorded.

Neuromuscular Function (Handgrip Strength): To assess neuromuscular function, a bilateral maximal voluntary contraction (MVC) handgrip test was used. Two digital calibrated hand dynamometers (DynX®, MD Systems, Reynoldsburg, OH, USA) were employed, equipment previously validated for reliability and accuracy [36]. Drivers performed the test while seated, simulating the driving position, with the torso upright, forearms supported, and elbows positioned slightly behind the edge of a table, holding a dynamometer in each hand. After standardized instructions, participants were asked to perform a simultaneous maximal contraction with both hands, squeezing the dynamometers as hard as possible (moving the thumbs toward the other fingers). Each assessment consisted of two maximal attempts of approximately three seconds, separated by about thirty seconds of rest; the maximal force value (in kg) achieved with each hand was used for analysis. Handgrip measurements were taken before the race/film (neuromuscular baseline) and repeated after the race/film, allowing the calculation of changes in maximal force as an indicator of peripheral muscular function and fatigue.

Cognitive-motor Performance: The BlazePod™ system (Play Coyotta Ltd., Tel Aviv-Yafo, Israel) was used to assess the drivers' cognitive-motor performance, focusing on response agility and inhibitory control. BlazePod consists of portable and interactive light sensors (pods) that can be configured in different spatial arrangements. In this study, six pods were used, placed on a table in a rectangular shape (three on each side) with 30 cm spacing between them.

The selected test lasted 30 s: drivers, positioned at the center of the setup, were instructed to touch only when the light appeared in the predefined target color (purple), ignoring signals of other colors (green or blue), which served as distractor stimuli. Each test session presented random and simultaneous stimuli across the pods, with only one pod lighting up in the correct color at any given time, thus requiring quick reaction and the ability to inhibit responses to irrelevant stimuli. This protocol can be conceptualized

as a discrimination visuomotor task involving selective attention and response inhibition, functionally analogous to go/no-go paradigms commonly used to assess inhibitory control in cognitive neuroscience [37].

The BlazePod™ application (version 4.0.3) automatically recorded the minimum, maximum, and average reaction times (in milliseconds) for correct touches, as well as the number of hits (touches on the correct pods) and inhibition errors (touches on incorrect colors). Before the first assessment, all participants completed a familiarization session with the BlazePod, repeating the test procedure until they felt comfortable, in order to minimize learning effects on performance.

This type of task is associated with executive performance abilities, namely inhibitory control and reaction time, and BlazePod has been shown to be a reliable tool for measuring response times in this context [15]. Cognitive-motor assessments were carried out before and after each condition (race and control), similarly to the other measurements.

Interventions and Procedure: The sequence of assessments was determined by the official event schedule: the control condition (film exposure, without exertion) took place before the start of on-track activities, while the race-condition assessments followed the event's regulatory timetable. For logistical reasons inherent to real competition, it was not possible to randomize or counterbalance the order of conditions. To mitigate learning effects, standardized familiarization with the cognitive-motor test (i.e., BlazePod) was implemented before valid data collection.

Race Viewing in Audiovisual Format: In the control condition, participants were exposed to an audiovisual stimulus with a duration equivalent to that of a karting race in order to reproduce the same time interval without the physical component of competition. For this purpose, a karting race video available online (YouTube—Champions of the Future, https://www.youtube.com/watch?v=NKGEax-cpHE (accessed on 8 March 2025) (18 min and 40 s duration time)) was presented and watched in full by the drivers in a controlled environment. This strategy aimed to standardize the exposure time and ensure that participants remained in a passive observation situation, allowing for direct comparison with the intervention condition (actual race).

Qualifying and Race: In the intervention condition, the drivers participated in the official races of the Portuguese Karting Championship—Toyota 2025, held at the Viana do Castelo Karting track. These events were organized in accordance with the International Sporting Code (ISC), the General Regulations for Motorsport and Karting (PGAK), the Specific Regulations for Karting (PEK), and the Supplementary Regulations of the event, homologated by the Portuguese Automobile and Karting Federation (FPAK).

The races took place on a circuit with a perimeter of 1117 m and a width of 10 m, run in a clockwise direction. According to the regulations, the number of laps varied by category. The X30 Junior and X30 Senior categories consisted of 18 laps in the qualifying heats and 20 laps in the final race. The X30 Super Shifter category included 15 laps in the qualifying heats and 18 laps in the final race. Because lap counts differed by category, total race duration varied accordingly.

Each competitive event was composed of different phases, namely the timed practice sessions (qualifying), qualifying heats, and the final race. In the timed practice sessions, drivers competed for the best lap time, which determined the starting grid for the heats. The qualifying heats were intermediate races contested by all participants, with the sum of the results determining the starting position for the final. The final race was the main event, with a longer duration than the heats, where the finishing order determined the final classification and the allocation of points.

During these races, drivers were subjected to the real demands of competition, including continuous physical effort to control the kart without power steering while resisting

lateral G-forces and intense chassis vibrations, environmental exposure with high temperatures and mandatory protective equipment (homologated suit, full-face helmet, gloves, rib protector, and boots), as well as psychological pressure resulting from close proximity between karts, position battles, and the need to make split-second decisions. All participants raced under official championship conditions, which inherently demanded maximal competitive engagement.

This competitive framework ensured that the intervention condition authentically reproduced the environment of an official race, allowing the evaluation of the acute effects of competition on the physiological (HR and HRV), neuromuscular (function and handgrip strength), and cognitive (reaction times and cognitive-motor performance) variables under study.

Schedule and Assessments Schedule: The official event schedule ran from 9:00 a.m. to 6:00 p.m. Measurements were performed at four distinct time points: before the control condition (video viewing), after the control condition, after the qualifying session, and after the final race.

To ensure consistency, all assessments were temporally standardized: 30 min before each competitive moment, at rest, and 10 min after the end of the respective race or condition, during the immediate recovery phase. This procedure ensured comparability between conditions while respecting the official competition calendar across the different categories.

Participants Preparation: Before data collection, all participants received a familiarization session with the instruments used, including the handgrip dynamometer, the perceived fatigue questionnaire, and the BlazePod® system for reaction time assessment. Each driver performed preliminary trials until they fully understood the procedure, ensuring consistency in responses and minimizing learning effects.

Additionally, participants were instructed to maintain a standardized posture during the use of the equipment (seated, with upper limbs supported at a 90° angle in the case of the MVC).

Environmental and Logistical Conditions: According to real-time meteorological information provided by specialized platforms, on 8 and 9 March 2025, in Viana do Castelo, mild temperatures were recorded (minimums of 9 °C and maximums between 13 and 14 °C), accompanied by cumulative precipitation exceeding 20 mm on each day. These conditions kept the track permanently wet and required the use of rain tires in all sessions of the event. Thermal comfort/discomfort was not quantified; thus, its potential contribution to perceived fatigue and HRV responses cannot be excluded.

Statistical Analysis: Data analysis was performed using non-parametric methods to account for repeated measures and potential violations of normality assumptions. Changes across multiple time points (pre-control, post-control, post-qualifying, and post-race) were assessed using the Friedman test. When a significant main effect was detected, post hoc pairwise comparisons were conducted using Wilcoxon signed-rank tests with Bonferroni correction for multiple comparisons. Exploratory data analysis was conducted, and data normality was assessed using the Shapiro–Wilk test.

Descriptive statistics are presented as median [percentile 25–75], and statistical significance was set at $p < 0.05$. All analyses were performed using IBM SPSS Statistics (version 29.0.1.0).

5. Conclusions

This exploratory field study, conducted during an official karting event, showed that competition was associated with reductions in time-domain HRV indices (rMSSD and SDNN), consistent with heightened autonomic activation during qualifying and racing. In contrast, perceived fatigue, handgrip-based neuromuscular function, and visuomotor

reaction-time outcomes did not show clear impairments within the immediate post-session window, while accuracy was higher on the second day of competition. These findings suggest that, under the specific demands of this event, increased physiological activation did not translate into detectable neuromuscular or cognitive-motor decrements.

From a practical perspective, monitoring HRV before and after competitive sessions may help characterize activation levels and support recovery-related decisions, particularly when subjective fatigue measures show limited immediate sensitivity. However, the small and heterogeneous sample, the fixed condition order imposed by the official race schedule, and the use of short PPG-based HRV recordings limit generalizability. Importantly, the observed reductions in HRV should be interpreted as markers of acute autonomic stress and competitive arousal rather than as indicators of fatigue per se.

Recent advances in wearable ECG-based technologies, including lightweight adhesive patches, increasingly allow reliable cardiac monitoring in field-based and competitive motorsport settings and should be considered in future studies. Larger, more homogeneous samples and ECG-based monitoring are warranted to further clarify the relationships between physiological activation, fatigue, and performance in karting.

The present findings should be interpreted cautiously and considered preliminary.

Author Contributions: Conceptualization and study design were led by D.V.L. and A.J.A. with input from all authors (M.C.P. and F.Y.N.). M.C.P., D.V.L. and A.J.A. conducted data collection and preliminary analysis. M.C.P. performed the detailed data analysis and interpretation and drafted the initial manuscript, which was critically revised by all authors. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the University of Maia (Comissão de Ética da Universidade da Maia), approval code: N.º 277/2025, approval date: 18 March 2025, under the project entitled *"Efeitos da Corrida na variabilidade da frequência da frequência cardíaca, fadiga e desempenho cognitive-motor em pilotos de automobilismo."*.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study. Ethical approval and consent procedures were reviewed and approved by the Ethics Committee of the University of Maia (project reference: 277/2025).

Data Availability Statement: All aggregated data are included in this article. Raw data are available from the corresponding author upon reasonable request, subject to ethical approval.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ECG	Electrocardiogram
FPAK	Portuguese Automobile and Karting Federation
G-forces	Gravitational Forces
HR	Heart Rate
HRV	Heart Rate Variability
HRV4Training	Heart Rate Variability for Training (Mobile Application)
ISC	International Sporting Code

LMM	Linear Mixed Model
M1, M2, M3, M4	Measurement Moments 1–4
MVC	Maximal Voluntary Contraction
PEK	Specific Regulations for Karting *(Prescrições Específicas de Karting)*
PGAK	General Regulations for Motorsport and Karting *(Prescrições Gerais de Automobilismo e Karting)*
PPG	Photoplethysmography
rMSSD	Root Mean Square of Successive Differences Between Normal RR Intervals
ROF	Rating of Fatigue Scale
RR	R–R Interval
SDNN	Standard Deviation of All Normal RR Intervals
UL	Upper Limbs

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